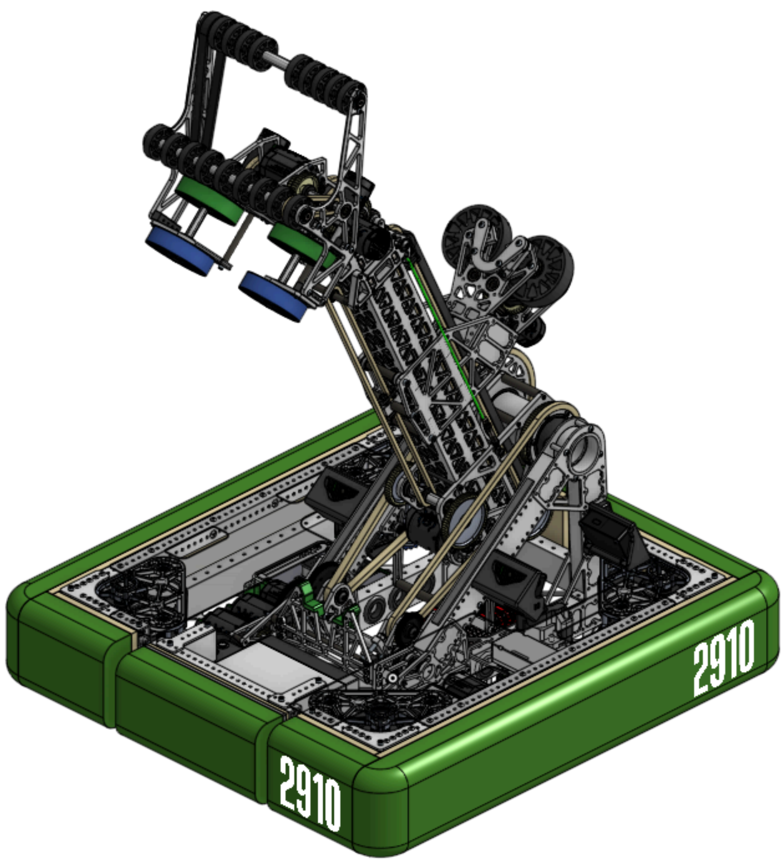
2910 JACK IN THE BOT

SPECTRE



Tech Brief 2025

2910 Goals and Guiding Principles

Our Mission:

Inspire through excellence and engage through “interesting fun stuff.”

2025 Reefscape Season Competitive Goal:

Win the World Championship

Strategy Philosophy - Guiding Principles for Strategy Priorities:

- Prioritize Einstein-level gameplay
- Prioritize playoffs gameplay
- Plan for alliance-level autonomous
- Minimize cycle times

Design Philosophy - Guiding Principles for Robot Architecture:

- "Put the thing on the thing" - Adam Freeman (67)
- "Do less, better" - Ty Holtzmann (2056)
- The “wrong” choice early is better than the “right” choice too late - Adam Heard (581, fmr 973)
- It doesn't really matter what it is, as long as it's fast

Design Objectives - Metrics for Design Success:

- **Reliable** - designed for modern FRC environment without losing performance
- **Fast** - 0.25 - 0.5 seconds for all movements and scoring actions
- **Stable** - CG near center and below top of bumpers in transit config
- **Aggressively Simple** - scrutinize tactical features
- **Lightweight** - 90lbs w/o battery and bumpers
- **Familiar** - Use mechanisms we have experience with

We expect not to always be able to meet all of our design objectives simultaneously - each game will require discussion and buy-in of which objectives we choose not to meet. For 2025 Reefscape, we chose to ignore the “Agressively Simple” and “Lightweight” requirements, based on our game analysis.

Strategic Priorities and Subsystem Requirements

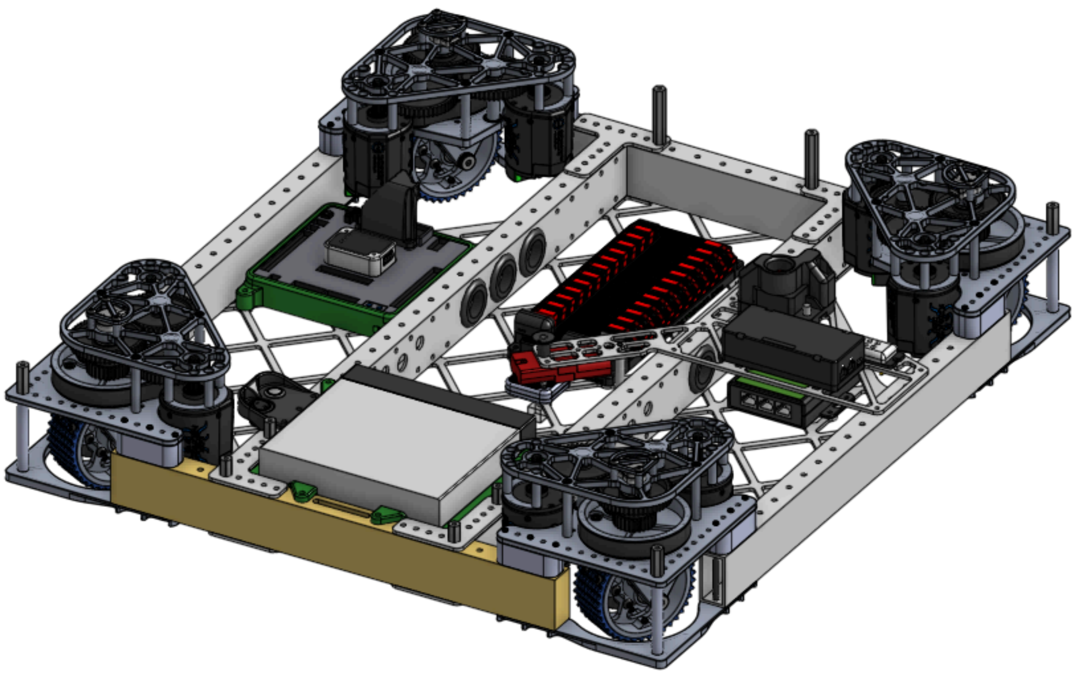
Must (core robot functions)	Could (pursue if convenient)	Won't (no design effort)
Drive	Score Algae in processor	Shallow Cage Climb
Deep Cage climb		Intake Algae from HP at Processor
Score Coral L1-4		Intake Coral off the Reef
Intake Coral from ground		Buddy Climb
Intake Coral from Coral Station		Get Buddy Climbed
Intake Coral from Mark		Control both Algae and Coral at same time
Score Algae in Net		Intake Algae and score Coral in the same motion
Intake Algae from floor		
Intake Algae from Reef		

General	Drive	Coral Intake	Algae Intake	L1-L4 Scorer	Algae Net	Deep cage	Automation
CG below 7 inches in travel configuration	Optimize for sprint distance of station to reef	Touch it, own it	Touch it, own it	Must be the same mechanism for all levels	>95% success rate in scoring pieces	<2 seconds to climb	Has autos
Hold preload game piece	Omnidirectional	Intake from the ground	Intake from reef	<1 second from Coral controlled to scoring position	<1 second from piece controlled to scoring position	<3 seconds to line up for climb	Automatic alignment to score
All crucial electronics are easily accessible (can be reached within <30 secs)		Intake from the station	Intake from the ground	<1 second from scoring position to scored	<1 second from scoring position to scored	Touch it, own in <0.5 seconds	Feedback for whether or not we have a piece and which one
Able to drive through coral and algae		Withstand load of another robot pinning/hitting our robot while it is deployed	Withstand load of another robot pinning/hitting our robot while it is deployed	Robot is stable while extending and scoring, wheels stay on the ground	Score in net when other algae are already in the net	Can control slightly swinging cage	Automated climb
Start in legal configuration		Intake from any orientation	Intake while algae is against the wall	>95% success rate in scoring pieces	Does not knock the other algae out of the net	Stay climbed for 4 seconds after robot is disabled even while the cage is slightly swinging	
Robust such that contact from Cages does not damage the robot		Intake while coral is against the wall	Intake while the robot is moving at 50% velocity	Able to be scored even when there is a coral diameter between the robot and the reef	Don't touch the net	Must be removable without enabling	
Anything that goes outside the bumpers is replaceable within 8 mins		Intake while the robot is moving at 50% velocity	Intake stows inside bumpers	Use 30 deg triangle to wheelbase to determine CG requirement while extended			
Maintain control of game pieces through collisions		Intake is on opposite side of scoring					
No jams in any system that interfaces with game pieces		Intake stows inside bumpers					

2910
2910

Drivebase

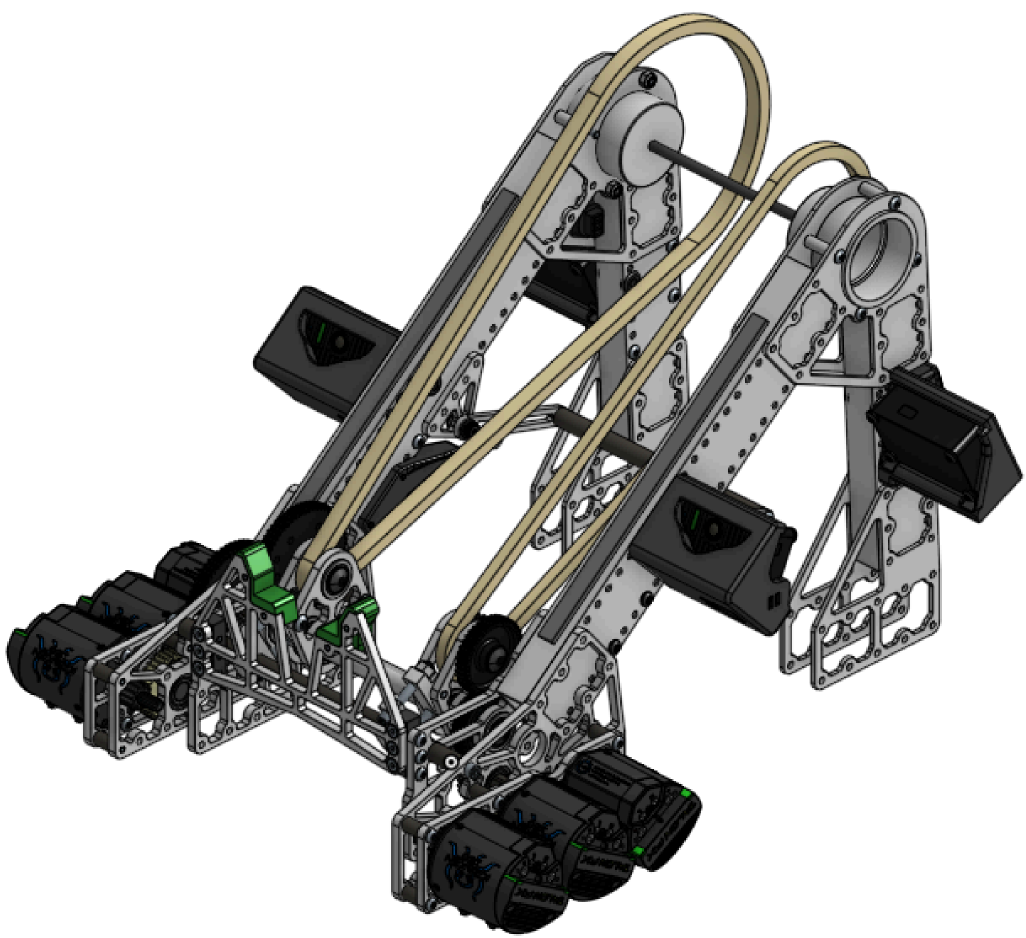
The Drivebase is the base structure of the robot, integrating the swerve drive modules, electronics and providing mounting provisions for the pivot and arm. It is designed to be fast, maneuverable and robust enough to survive a season of high speed collisions



- 2''x1'' chassis with SDS lowering kits to lower CG
    - 10lb solid brass ballast bar to bias CG forward and counteract arm motion
    - All major electronic modules mounted on the drivebase
    - Battery and RoboRIO mounted with custom 3D Printed TPU covers for FOD protection and shock isolation
- Mk4i modules for Districts and PNW District Championship
    - Swerve drive base offers speed and maneuverability
    - L2 gearing optimized for short cycles
    - 8x Kraken X60 Motors for steering and drive
- Mk5 prototype modules for World Championship
    - 4x Kraken X60 Motors for drive
    - 4x Kraken X44 Motors for steering
    - Fully enclosed gears, 2.25" wide wheels
    - Additional brass bar ballast inserted in rear chassis frame to lower CG from weight saved with Mk5 conversion (~5lbs)

Pivot

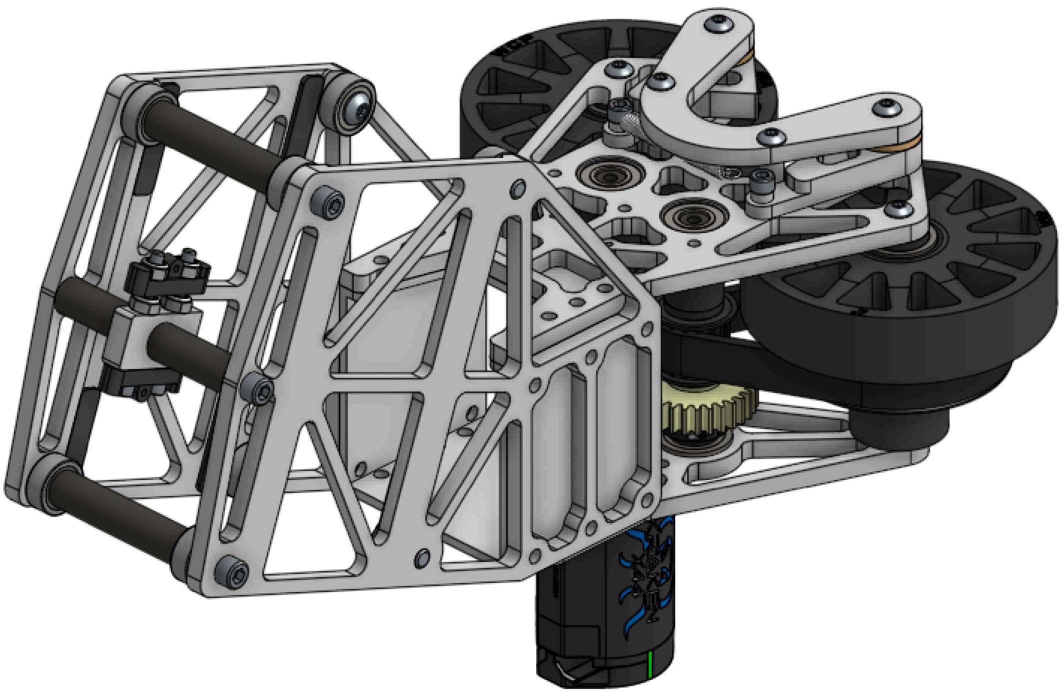
The pivot is a compact, highly integrated design that combines the gearbox that drives the pivot with the gearbox that drives the arm extension. The pivot provides a dead axle for the arm to rotate around, and a cradle to soften impacts and provide a hard stop to zero the arm sensors.



- 2 practically identical but functionally different gearboxes
    - Driven by 3x Kraken X60 for each gearbox
    - Pivot gearbox is driven at a 120.83:1 gear ratio with a #35 chain powering the final sprocket reduction
    - Arm extension gearbox is driven at a 6.28:1 gear ratio with a #25 chain powering the final sprocket reduction
- 4 Limelight mounts to assist with automated alignment to the reef from the front and back
- Pivot dead axle increases stiffness, is held together with 2 end plugs, and allows the arm to be removed with one bolt
- Kickstand holds arm in starting configuration while disabled

Climb

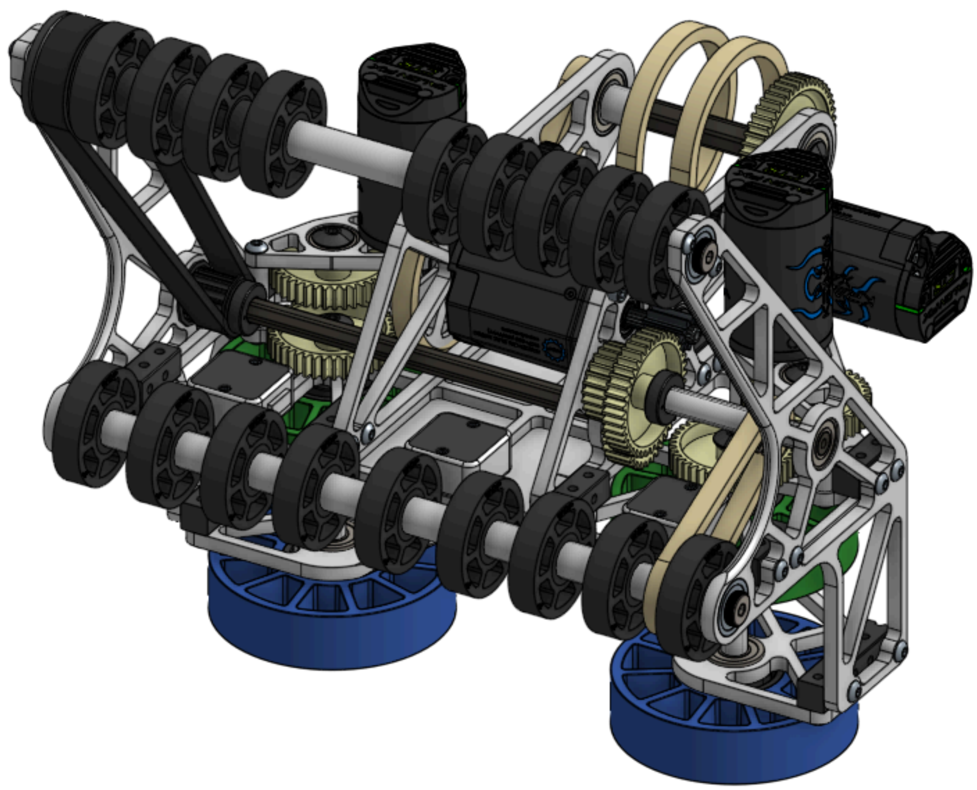
The climber was designed to be able to climb while gripping a single bar of the Deep Cage with a powered collector, reducing the need for precise alignment. It is mounted to a carriage which travels along the outside of the arm, allowing us to manipulate the anchor chain position for a balanced climb. The main lifting motion is accomplished by the arm shoulder, reducing complexity and weight compared to a separate subsystem.



- “Touch-It, Own-It” grabber intakes one of the cage bars in between the two wheels
    - Powered by 1x Kraken X44
- Passive spring-loaded latches secure cage bar for climbing, lower V wedge provides reaction point to rotate the cage
- Lifting motion is accomplished by pivoting the arm down to rotate the cage, and sliding the climber carriage along the arm to balance the robot CG below the anchor
    - Carriage motion powered by 1x Kraken X60
- The arm extension is used to hook into the front bumper frame to lock the arm

Intake and Wrist

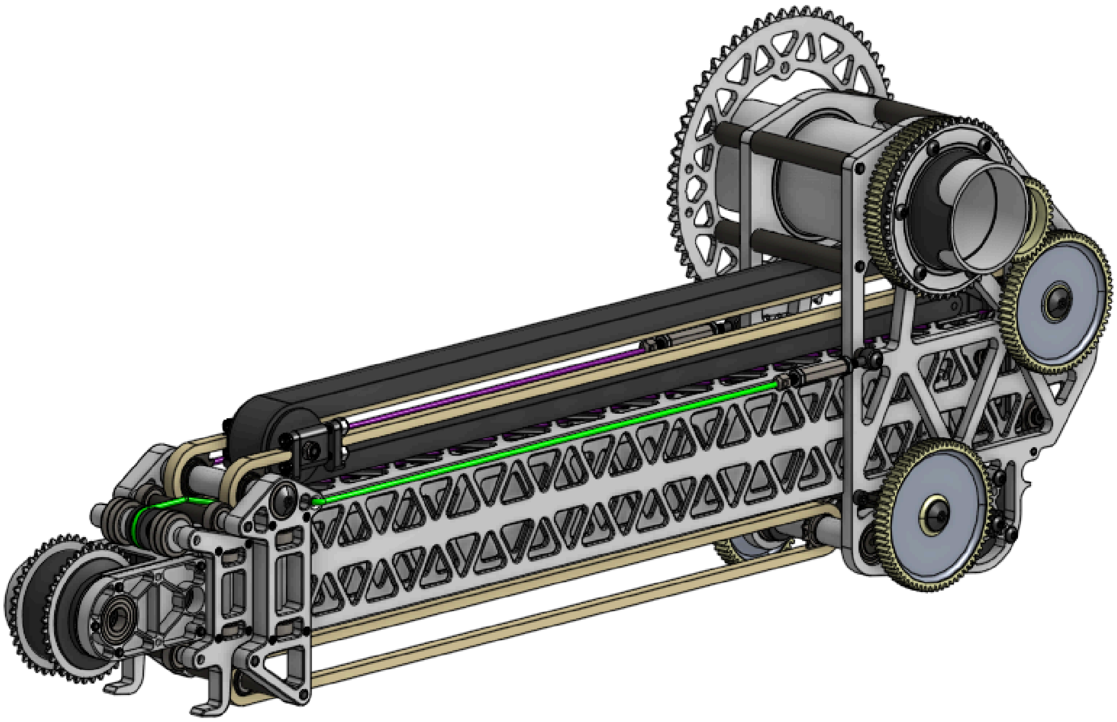
The intake picks up Coral from the ground and Algae from the ground and Reef with a deployable intake system. The intake also works as the scoring mechanism to score the two game pieces at all levels, forwards and backwards with the wrist.



- Coral Intake
    - Has four vertical wheels and a set of horizontal wheels to collect and orient Coral quickly either lengthwise for L2-L4 or horizontal for L1
    - Pass-through intake allows scoring in either direction
    - 4 CANRange sensors detect Coral and index it for scoring front, back or sideways
    - 1x Kraken X60 for horizontal rollers, 2x Kraken X44 for vertical wheels
- Algae Intake
    - Horizontal wheels from coral intake plus upper compliant wheel roller grab Algae
- Wrist uses reverted chain reduction for compact and light motion
    - Over 180* of motion powered by 1x Kraken X44

Arm

The Arm pivots and extends allowing us to move our intake to score Coral on all levels of the reef and score Algae in the net and processor. The pivoting motion of the arm is also used to climb and the extension used to lock into the bumpers so we stay climbed after the match. The offset pivot enables a stowed configuration with very low CG.



- Extension
    - Powered by the extension gearbox on the A frame
        - Arm extension sprocket is coaxial on the pivot dead axle
        - Gearbox power is transferred by a #25 chain
    - Two stage Cascade driven mechanism
        - First stage is driven by two #25 chains
        - First stage bearing blocks have hooks to lock the pivot during climb.
        - Second stage is driven by a 3mm UHMWPE string
- Pivot
    - Pivot dead axle is connected by a single bolt to facilitate arm maintenance
    - Pivot is powered by a gearbox on the A frame
        - Gearbox power is transferred by a #35 chain to the large sprocket
- Carriage gearbox
    - Gearbox and #25 chain on the bottom of the arm power Carriage motion for climber.

Software

- Multithreaded hardware signal processing integrated with AdvantageKit
- 4x Limelight 4's for robot relocalization
- Custom finite state machine architecture to coordinate robot actions
    - Xbox button presses push generic system state to superstructure which delegates actions to individual subsystems
- LEDs to indicate robot state to driver
- Fully automated coral scoring and reef algae pickups
    - Reef face to drive to, and scoring front/back is automatically determined
        - Pose-based: the robot selects the reef face that is closest to the odometry position of the drivebase, and selects the rotation that is closest to the face the reef
        - Rotation-based: the robot chooses between 2 reef faces that matches the gyroscope's closest 60-degree angle based on whether the front or back Limelights see the appropriate AprilTags for that side of the robot
    - Level and left/right determined by driver
        - L4, L3, L2, R4, R3, R2
        - (if holding horizontal coral) base L1, base R1, top L1, top R1
        - If picking up algae, L2 or L3 algae automatically determined by visible apriltag/selected reef face
    - Which Limelight pose to use automatically selected by whether is scoring front/back and left/right
    - Robot automatically drives to position and moves arm to the correct pose, and ejects once in tolerance:
        - Ejects piece (2-4)
        - Picks up algae, then drives out automatically